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BlockChain & DLT

EXPERIMENT NO - 8

Aim :- To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory

Q1. What is Blockchain?

Answer:

Blockchain is a distributed and decentralized digital ledger that records transactions across a network of computers (nodes). Each transaction is stored in a block, and blocks are linked together in chronological order, forming a chain. Once data is added, it cannot be altered, which ensures immutability. Because the ledger is shared among multiple participants, it provides transparency and eliminates the need for a central authority. This makes blockchain highly suitable for applications such as financial systems, supply chain management, healthcare records, and voting, where trust, security, and data integrity are essential.

Q2. What is Ganache?

Answer:

Ganache is a personal blockchain environment designed specifically for Ethereum developers. It allows developers to create a local blockchain on their computer, complete with pre-configured accounts and test Ether. This makes it possible to deploy, test, and debug smart contracts without spending real cryptocurrency. Ganache also provides features like transaction history, block details, and gas usage statistics, which help developers understand how their contracts behave in a real blockchain setting. By simulating the Ethereum network locally, Ganache speeds up the development cycle and reduces the risks of deploying untested contracts on the main network.

Q3. What is the role of MetaMask in this experiment?

Answer:

MetaMask is a digital wallet and browser extension that connects users to blockchain networks. In this experiment, MetaMask acts as a bridge between the local Ganache blockchain and the Remix IDE. It allows users to:

- Manage multiple Ethereum accounts.
- Sign and send transactions securely.
- Interact with deployed smart contracts directly from the browser.

By integrating with Ganache, MetaMask enables developers to test contracts in a realistic environment while still using familiar wallet functionalities. It ensures that transactions are properly authorized and provides a user-friendly interface for blockchain interaction.

Q4. How is Blockchain used in charity donation systems?

Answer:

Blockchain technology can transform charity donation systems by making them transparent, traceable, and trustworthy. Every donation is recorded on the blockchain, creating a permanent and verifiable record. Donors can track how their funds are being used, ensuring that money reaches the intended recipients. This reduces the risk of fraud, mismanagement, or misuse of funds. Additionally, blockchain allows for real-time reporting, meaning charities can show donors exactly where contributions are going. This level of accountability builds trust between organizations and donors, encouraging more people to contribute.

Q5. What is the role of Smart Contracts in charity donations?

Answer:

Smart contracts play a crucial role in automating the donation process. They are self-executing programs stored on the blockchain that run when predefined conditions are met. For example, a smart contract can be programmed to release funds only when a charity provides proof of completing a project. This ensures that donations are used exactly as intended. Smart contracts eliminate the need for intermediaries, reduce administrative costs, and guarantee that transactions are secure and transparent. By enforcing rules automatically, they make charity systems more efficient and reliable.

Q6. Why is Blockchain beneficial for charity systems?

Answer:

Blockchain offers several benefits for charity systems:

- **Transparency:** Donors can verify every transaction and see how funds are allocated.
- **Security:** Data stored on the blockchain cannot be tampered with, reducing fraud.
- **Accountability:** Charities are held responsible for how they use donations.
- **Cost reduction:** By removing intermediaries, administrative expenses are minimized.
- **Global reach:** Blockchain enables cross-border donations without relying on banks or third parties.

Together, these advantages make blockchain a powerful tool for building trust and efficiency in charitable organizations.

Q7. What are the steps involved in connecting Ganache with MetaMask and Remix IDE?

Answer:

The process of connecting Ganache with MetaMask and Remix IDE involves several steps:

1. Install Ganache on your computer to create a local blockchain environment.
2. Open Ganache and note down the RPC server details (usually <http://127.0.0.1:7545>).
3. Import Ganache accounts into MetaMask using the private keys provided by Ganache. This gives MetaMask access to test Ether.
4. Configure MetaMask to connect to the local Ganache network by adding a custom RPC network.
5. Open Remix IDE in the browser and select MetaMask as the environment for deploying contracts.
6. Deploy and interact with smart contracts through Remix, with MetaMask handling transaction signing and Ganache simulating the blockchain.

This setup allows developers to test contracts in a safe, controlled environment before deploying them to a live blockchain.

Implementation :-

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
```

```
contract CharityDonation {

    address public owner;
    uint public totalDonations;

    struct Donation {
        address donor;
        uint amount;
        uint timestamp;
    }

    Donation[] public donations;

    event DonationReceived(address donor, uint amount);
    event FundsWithdrawn(address owner, uint amount);

    constructor() {
        owner = msg.sender;
    }
}
```

```

// Function to donate Ether
function donate() public payable {
    require(msg.value > 0, "Donation must be greater than zero");

    donations.push(Donation(msg.sender, msg.value, block.timestamp));
    totalDonations += msg.value;

    emit DonationReceived(msg.sender, msg.value);
}

// Function to check contract balance
function getBalance() public view returns (uint) {
    return address(this).balance;
}

// Only owner can withdraw funds
function withdraw(uint amount) public {
    require(msg.sender == owner, "Only owner can withdraw funds");
    require(amount <= address(this).balance, "Insufficient balance");

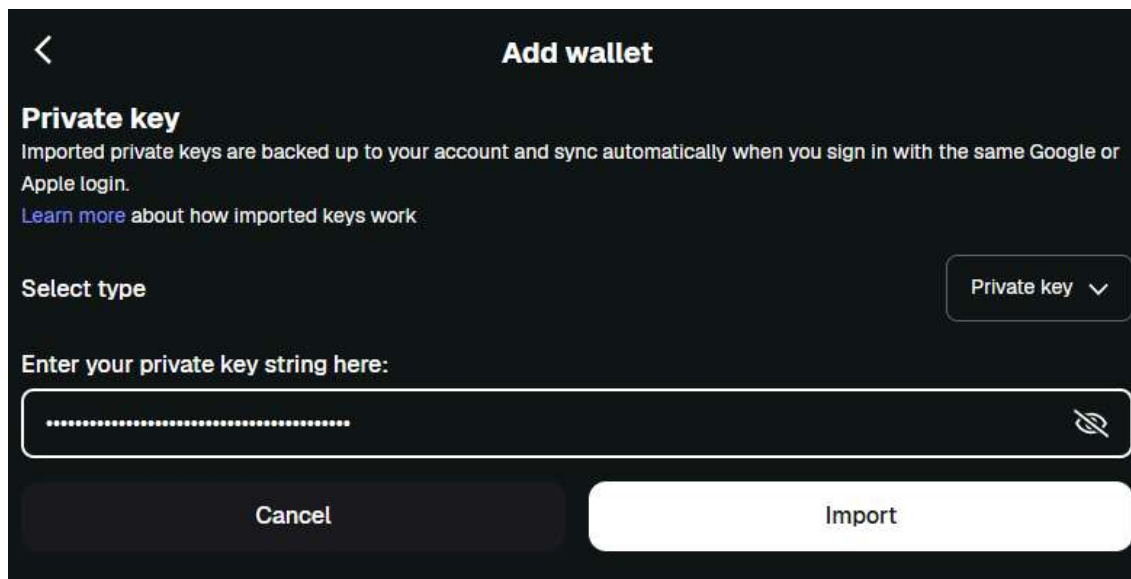
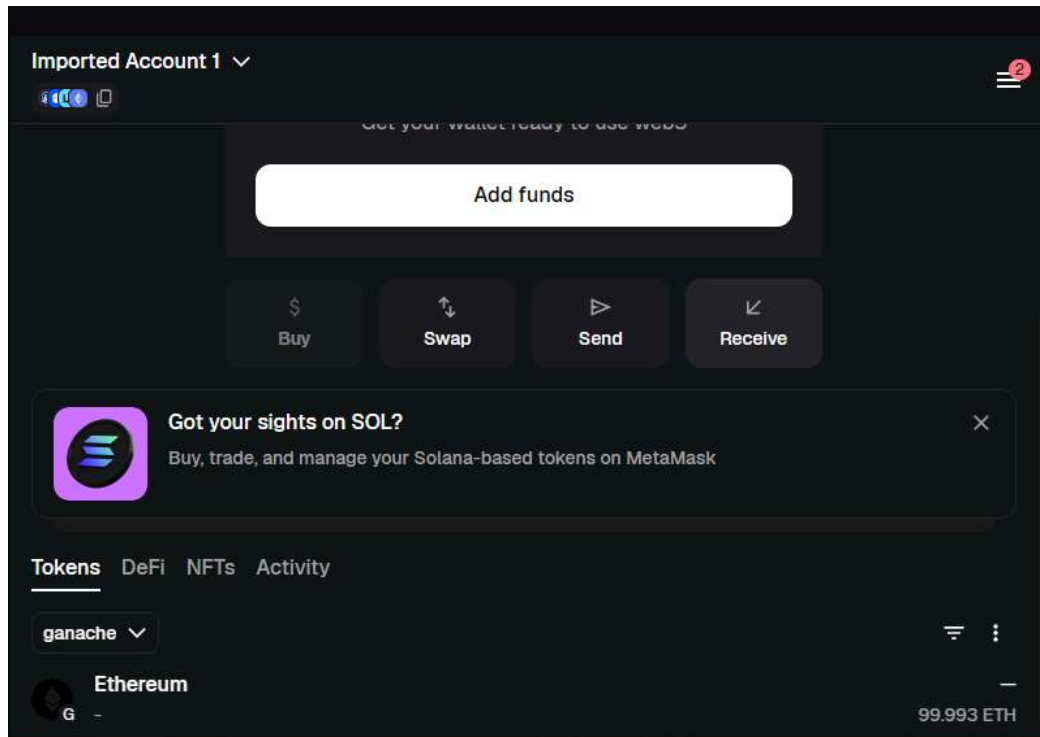
    payable(owner).transfer(amount);

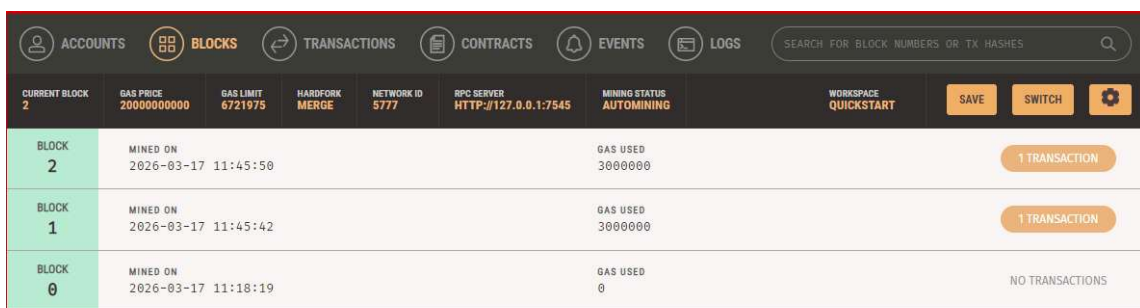
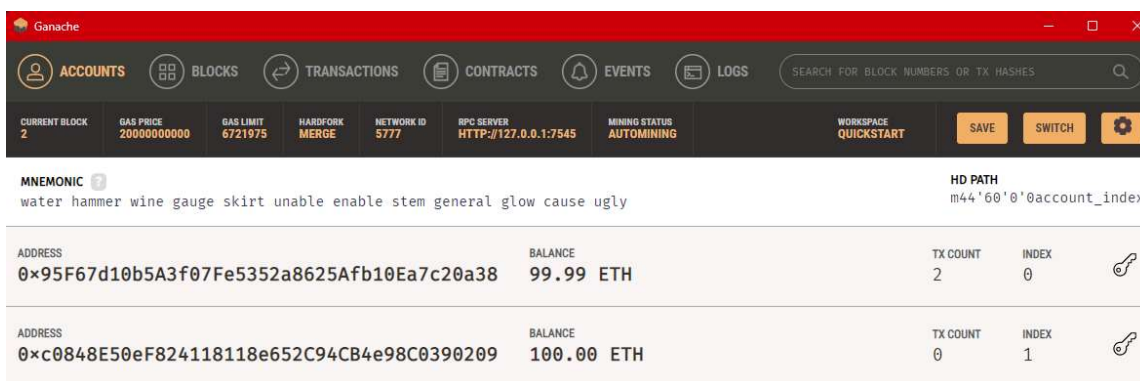
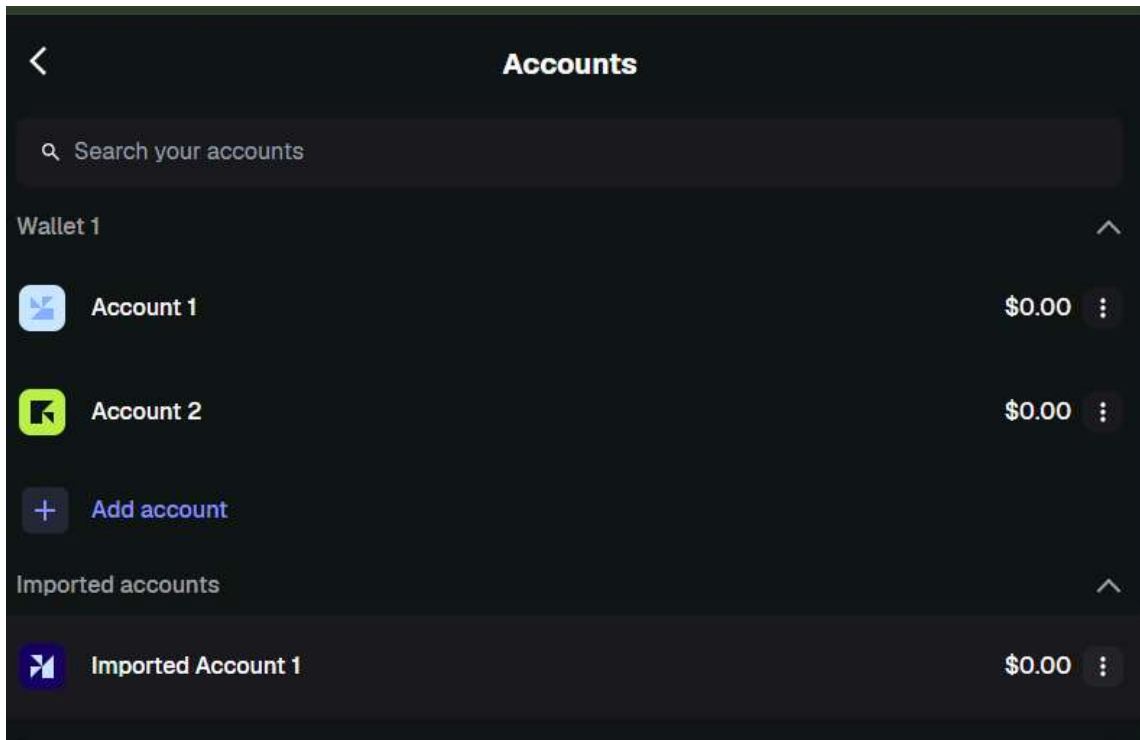
    emit FundsWithdrawn(owner, amount);
}

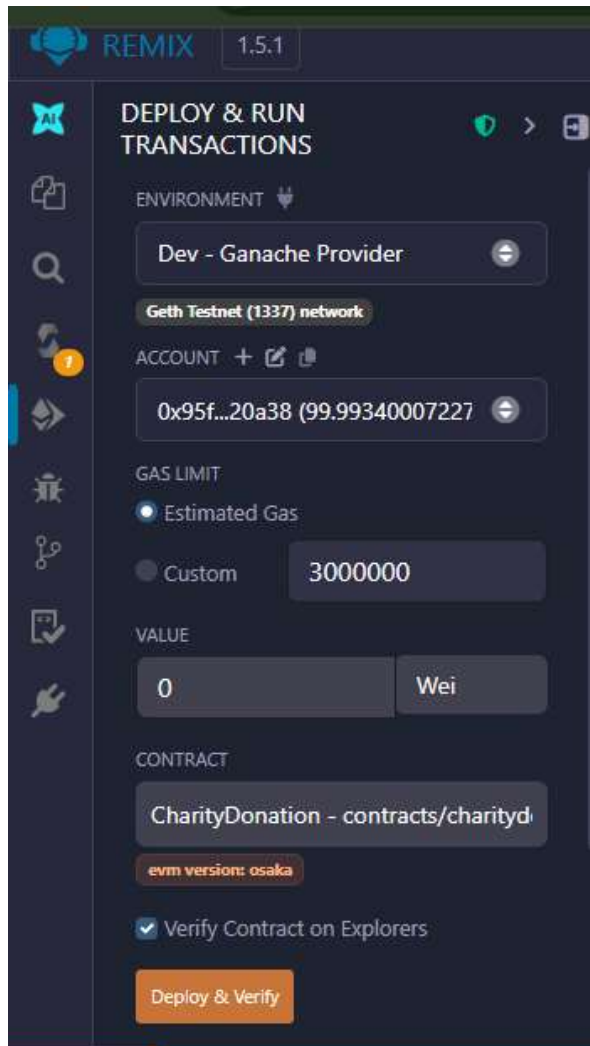
// View number of donations
function getDonationCount() public view returns (uint) {
    return donations.length;
}
}
Metamask

```

Output:







Conclusion :-

This experiment demonstrated the implementation of a blockchain-based environment using Ganache, MetaMask, and Remix IDE. A smart contract was successfully created, deployed, and tested using Ganache accounts integrated with MetaMask. The experiment also highlighted the potential of blockchain technology in enhancing charity donation systems by ensuring transparency, security, and traceability of transactions. Overall, it provided practical knowledge of executing and monitoring blockchain transactions in a controlled environment.